Fasteners – Torquing

1. General

This standard specifies the procedure and requirements for torquing bolts, nuts and screws (excluding fluid line fittings), which have torque values given in Maintenance Practices.

2. Materials and Equipment

A. Materials:

- MIL–PRF–83483 anti-seize thread compound
- Tamper proof sealant (putty), white, pink or red – Organic Products Co. F-900 or #3M EC1252. Refer to appropriate MSDS for the storage life of tamper proof sealant

B. Equipment:

- Torque wrenches
- Sockets and adapters as required

3. Requirements

A. Bolts, nuts and screws must only be torqued when specified in Maintenance Practices.

B. Lubricating and anti-seize compounds must not be used on threaded fasteners unless such preparation is specifically called up.

C. All paint, dirt and corrosion must be removed from threads prior to mating up and torquing. Mating parts must be smooth and clean.

D. Threaded fasteners must engage freely.

E. The installed bolt must not engage imperfect threads of tapped blind holes, and the threads of tapped holes must not engage the run-out of the bolt or screw threads.

F. Bolts, nuts and washers must seat squarely with the mating surfaces.

4. Torquing

A. Installed fasteners must be torqued to values specified in Maintenance Practices.

B. Wherever possible, in a bolt-nut or screw-nut combination, the bolt or screw must be held stationary and the nut turned.

C. When it is necessary to tighten a fastener from the bolt or screw side, the torque value must approach the high side of the specified range.

D. Whenever torqued fasteners are to be secured by means of cotter pins or lockwire, the low side of the torque range must be used for tightening. If necessary, tighten the fastener so that the next slot aligns with the hole, provided the maximum torque value is not exceeded.

E. When the maximum torque value is reached and the slot in the nut does not line up with the hole in the bolt, the nut and washer must be changed.

F. When installation of a part involves tightening a group of more than three fasteners, pre-torque all fasteners to 75% of final torque value in the following sequence, as applicable:

1) For relatively solid parts where length to width ratio of fastener pattern is less than two, pre-torque in a criss-cross order.

2) When length to width ratio of fastener pattern is two or more, or when the flatnose of thin parts or gaskets may be affected by the torquing sequence, pre-torquing must be carried out starting from the center of the pattern and working outwards to the ends.

G. After pre-torquing, torque all fasteners to final torque value in same sequence as used in pre-torquing.

5. Overtorqued Fasteners

A. Fasteners which have been tightened beyond the maximum specified torque value must be removed, rendered unserviceable, and scrapped.

B. Overtorqued fasteners must not be backed off and retorqued to the correct value.

C. In the case of overtorqued bolt-nut or screw-nut fasteners, both bolt or screw and nut must be discarded.

6. Retorquing

A. Where it is necessary to retorque a fastener assembly, the nut must be backed off one full turn and retightened to the specified value.

B. The bolt or screw must not be allowed to turn throughout the operation, unless the installation requires the torque to be applied to the bolt or screw in the first instance.

7. Use of Torque Wrenches and Adapters

A. Before use, pre-set click type torque wrenches to the specified value as follows.

1) Release the locking mechanism located on the wrench handle.

2) Adjust the wrench by rotating the handle until the desired setting is reached, as indicated by the graduated scale.

3) Engage the locking mechanism and visually re-check the torque setting.

B. When torquing fasteners, the torque wrench drive must be held squarely to the mating surface receiving the fastener.

C. A slow steady motion must be applied to torque wrenches when tightening a fastener.

D. Torque wrenches must be checked for accuracy before use.

E. When using offset adapters, the adapter must be in line with the torque wrench arm.

F. Allowance for the offset adapter must be made to calculate torque value in accordance with the following formula:

Dial Reading = (Specified Torque x L) ÷ (L + L1)

L = Distance in inches from mid point of grip to centerline of wrench drive.

L1 = Distance in inches from centerline of wrench drive to centerline of offset adapter drive.

8. Identification of Torqued Fasteners

A. All torqued fasteners should be identified after torquing.

B. Identification should consist of a stripe or blot of red lacquer.

C. Identification must be placed on the nut of any bolt-nut or screw-nut combination, whenever practicable, and mating surface, otherwise head of bolt or screw must be marked.

D. Where external appearance of aircraft may be affected, identification may be omitted.

E. If at any time misalignment or displacement of the identification stripe or bolt indicates relative movement of the fastener, the fastener must be retorqued in accordance with Para 6., and identification reapplied.

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